Semester VI **B. Tech CSE (AI)**

BCSAI 603: ADVANCED MACHINE LEARNING

3L + 0T + 0P + 3C **MM 100**

Unit 1: Introduction

Bayesian Machine Learning: Machine Learning Paradigms, types of machine learning approaches, delineating between supervised and unsupervised learning, and between discriminative and generative approaches.

Unit 2: Bayesian Modelling

Bayesian Modelling: assumptions and processes of constructing a Bayesian model, dependency relationships in Bayesian models,graphical models and probabilistic programming.

Unit 3: Bayesian Inference

Bayesian Inference: approaches for estimating Bayesian posteriors, marginal likelihoods, and expectations. Monte Carlo sampling, Markov chain Monte Carlo (MCMC) sampling and variational inference.

Unit 4: Natural Language Processing:

Challenge of Natural Language Processing (NLP), Embeddings : methods to create embeddings, disadvantages and advantages, Classification and neural networks: classification task, tasks arises in NLP problems, Language models.

Unit 5: Deep Learning

Motivation for deep learning, basic supervised classification task, optimizing logistic classifier using gradient descent, stochastic gradient descent, momentum, and adaptive sub-gradient method.

Text and Reference Books:

1. Jeff Heaton, Deep Learning and Neural Networks, Heaton Research Inc, 2015.
2. Ethem Alpaydin, Introduction to Machine Learning, Second Edition
3. Stephen Marsland, Machine Learning: An Algorithmic Perspective.
4. Tom Mitchell, Machine Learning